

Cyto-molecular Techniques

Done by

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Case 1

A 2-year-old child is brought by parents for global developmental delay. Parents report the child has mild hypotonia, delayed milestones (sat at 10 months, walking not yet independent), limited speech, and occasional brief “staring” episodes (no confirmed seizures).

On exam, the clinician notes dysmorphic facial features: prominent forehead, downslanting eyes, low-set ears, and a short upturned nose, plus clinodactyly. Growth parameters are normal; no major organ issues are found.

Family history: noncontributory; no consanguinity.

What genetic causes should be considered, and what initial investigations (e.g., genetic testing options) would you order next?

Case 2

A 45-year-old man presents with blood in stool and change in bowel habits for 2–3 months. Colonoscopy shows an ascending colon adenocarcinoma. Tumor pathology demonstrates loss of mismatch repair (MMR) protein expression (MMR-deficient). His mother had colon cancer at 48, and his maternal aunt had endometrial cancer at 52.

Identify the most likely diagnosis and suggest the next step in confirmation

What are the associated genes and mode of inheritance?

What is the recurrence risk in the next pregnancy?

Case 3

A 16-year-old presents with tall stature and long, thin limbs. He reports back pain and occasional palpitations. Ophthalmology evaluation revealed lens dislocation

On exam you note:

Arachnodactyly

Positive wrist and thumb signs

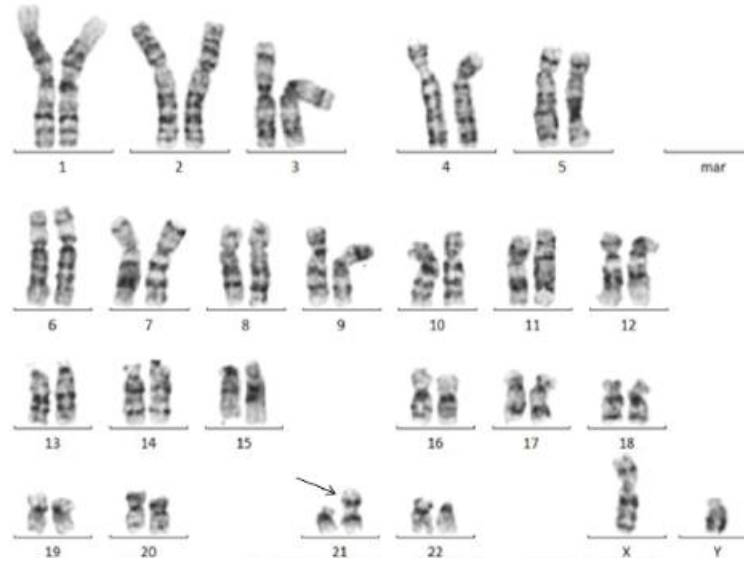
Scoliosis.

-Which genetic test would you like to order to confirm the most likely diagnosis in the single associated gene?

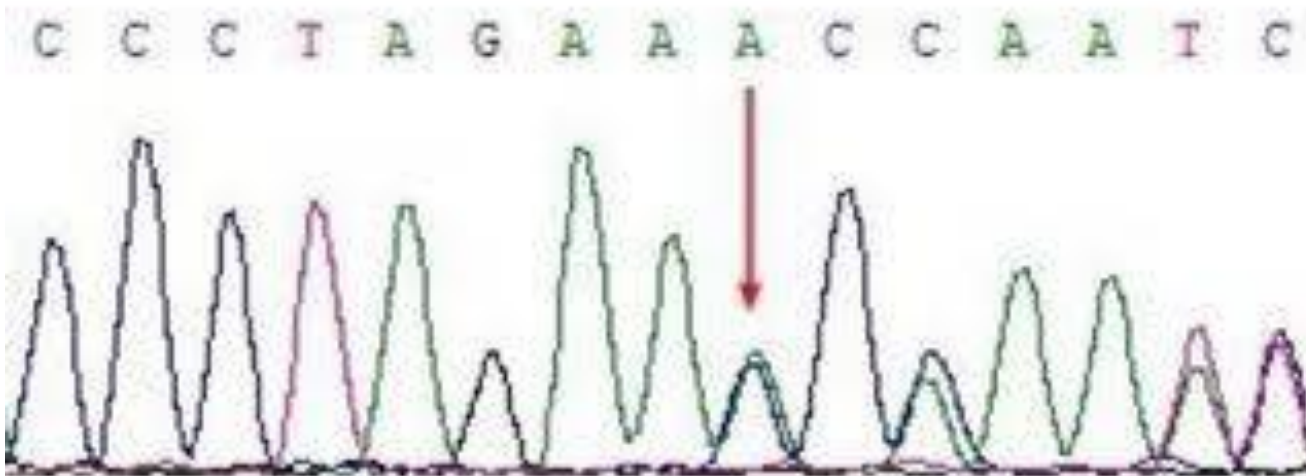
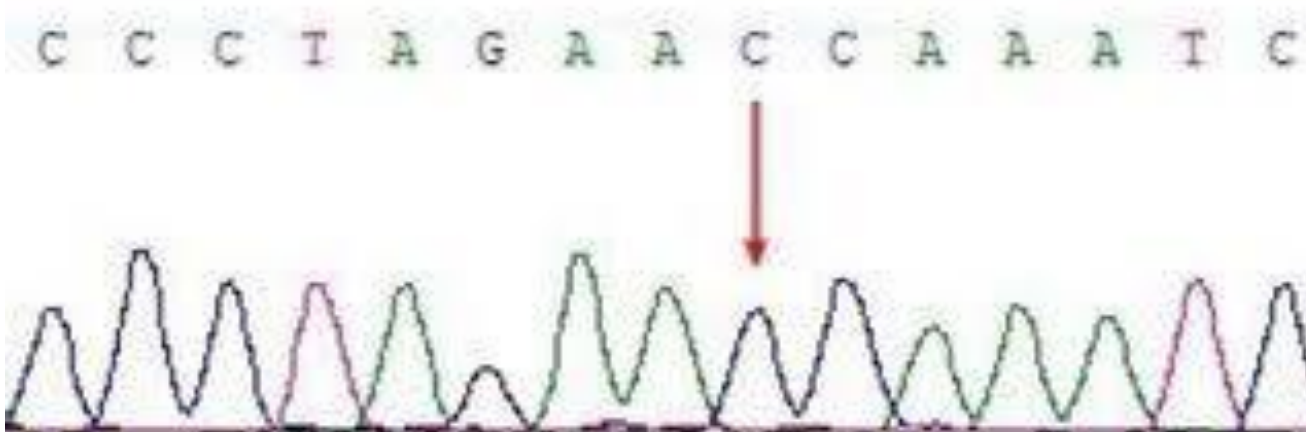
-If the genetic test that you ordered showed a negative result, what is your next step?

Case 4

- What is your next step?

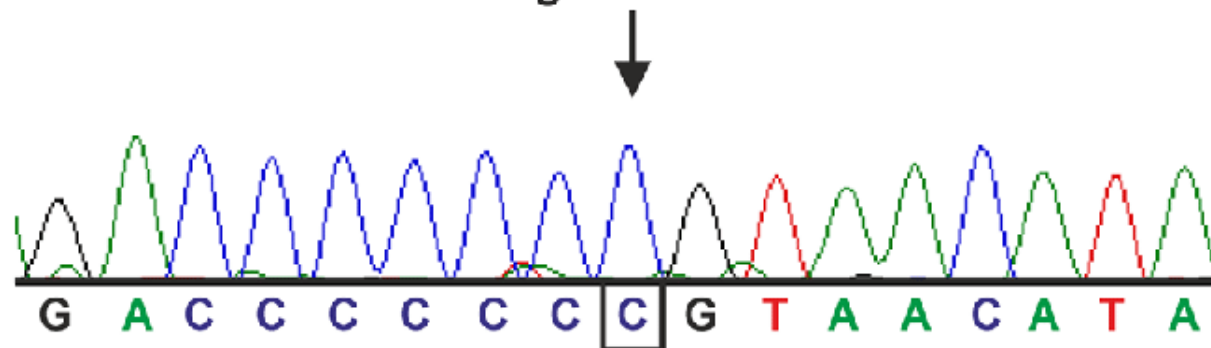


Insertion of A



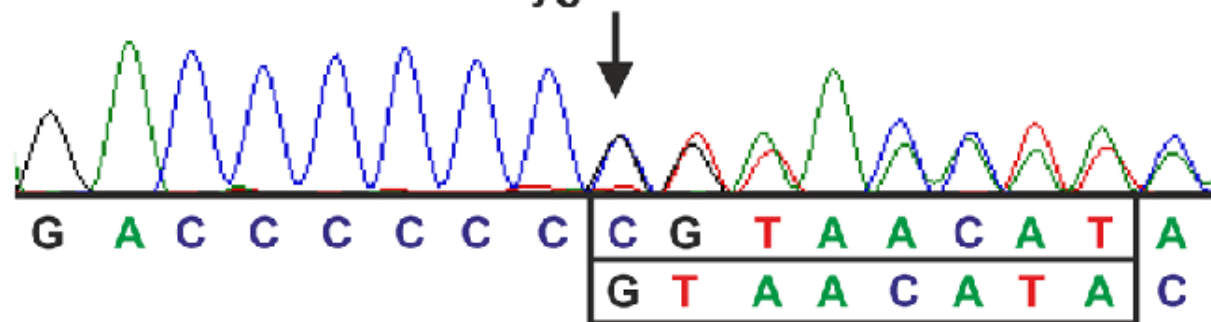
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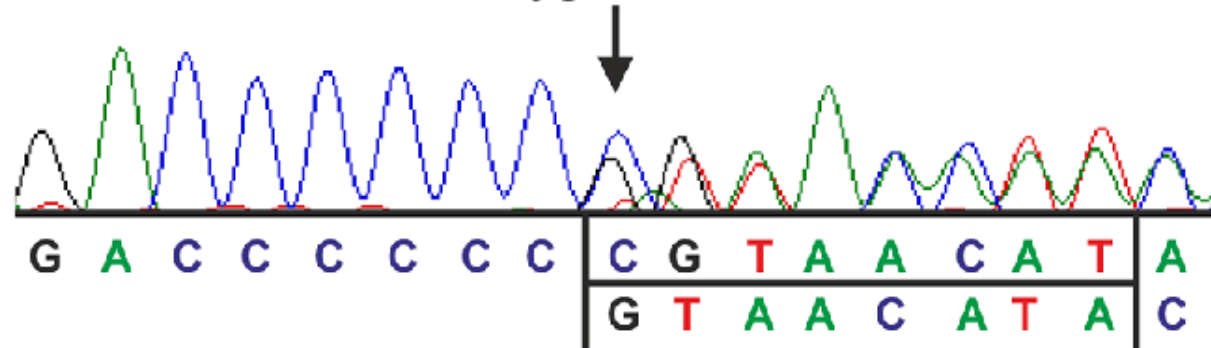
Mother

Heterozygous Insertion



Father

Heterozygous Insertion



من أشعار د. وسام عجيب

وما المرء إلا أخلوق
قد تبلغ به الآفاق
أو تجعله في قاع الأنفاق
فلا تكثر من الشقاق والنفاق
وكن بلسما في كل حال وترياق